

1. Write a C++ program to create a class for a student with a constructor and a destructor.

Code: #include <iostream>

using namespace std;

```
class Student{
public:
    Student(){
        cout<<"constructor called"<<endl;
    }

    ~Student(){
        cout<<"destructor called"<<endl;
    }
};

int main(){
    Student s;
    cout<<"student created object"<<endl;
    return 0;
}
```

SIMATS Online Programming Lab

V.Tejaswini
192424360RD

C++ PROGRAMMING

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 class Student{
4     public:
5     Student(){
6         cout<<"constructor called"<<endl;
7     }
8     ~Student(){
9         cout<<"destructor called"<<endl;
10    }
11 };
12 int main(){
13     Student s;
14     cout<<"student created object"<<endl;
15     return 0;
16 }
```

INPUT

Your INPUT goes here!

```
constructor called  
student created object  
destructor called
```

2. Write a C++ program to create a class for a book with a constructor and a destructor.

```
#include <iostream>  
using namespace std;  
  
class Book{  
public:  
    Book(){  
        cout<<"constructor called"<<endl;  
    }  
  
    ~Book(){  
        cout<<"destructor called"<<endl;  
    }  
};  
  
int main(){  
    Book b;  
    cout<<"book created object"<<endl;  
    return 0;  
}
```

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```
1 #include<iostream>
2 using namespace std;
3 class Book{
4     public:
5     Book(){
6         cout<<"constructor called "<<endl;
7     }
8     ~Book(){
9         cout<<"destructor called"<<endl;
10    }
11 };
12 int main(){
13     Book b;
14     cout<<"book object created"<<endl;
15 }
16
17
```

INPUT

Your INPUT goes here!

OUTPUT

```
constructor called
book created object
destructor called
```

3. Write a C++ program to create a class for a rectangle with a constructor and a destructor.

```
#include <iostream>
using namespace std;
```

```
class Rectangle{
public:
    Rectangle(){
        cout<<"constructor called"<<endl;
    }

    ~Rectangle(){
```

```

        cout<<"destructor called"<<endl;
    }
};

int main(){
    Rectangle r;
    cout<<"rectangle created object"<<endl;
    return 0;
}

```

The screenshot shows the SIMATS Online Programming Lab interface. At the top, there's a header with the lab name and a user profile for V. Tejaswini. Below the header, the 'PROGRAM EDITOR' section contains a C++ code snippet. To the right of the editor is an 'INPUT' field. Below the editor, the 'OUTPUT' section displays the results of running the code.

PROGRAM EDITOR

```

1 #include <iostream>
2 using namespace std;
3 class Rectangle{
4 public:
5     Rectangle()
6 {
7     cout<<"constructor called"<<endl;
8 }
9     ~Rectangle(){
10         cout<<"deconstructor called"<<endl;
11     }
12 };
13 int main(){
14     Rectangle r;
15     cout<<"rectangle created object"<<endl;
16 }
17
18
19
20

```

INPUT

Your INPUT goes here!

OUTPUT

```

constructor called
rectangle created object
deconstructor called

```

4. Write a C++ program to create a class for a car with a constructor and a destructor.

```
#include <iostream>
using namespace std;

class Car{
public:
    Car(){
        cout<<"constructor called"<<endl;
    }

    ~Car(){
        cout<<"destructor called"<<endl;
    }
};

int main(){
    Car c;
    cout<<"car created object"<<endl;
    return 0;
}
```

The screenshot displays the SIMATS Online Programming Lab interface. At the top, there is a dark blue header with the text "SIMATS Online Programming Lab" and a user profile icon for "V.Tejaswini" with ID "192424360RD". Below the header, a tab labeled "C++ PROGRAMMING" is active. The main area is divided into three sections: "PROGRAM EDITOR", "INPUT", and "OUTPUT". The "PROGRAM EDITOR" section contains a code editor with the following C++ code:

```
1 #include <iostream>
2 using namespace std;
3
4 class Car
5 {
6     public:
7     Car(){
8         cout<<"constructor called"<<endl;
9     }
10    ~Car(){
11        cout<<"deconstructor called"<<endl;
12    }
13 };
14 int main(){
15     Car c;
16     cout<<"Car  created object"<<endl;
17 }
18
19
20
21
```

The "INPUT" section on the right contains the text "Your INPUT goes here!". The "OUTPUT" section at the bottom is currently empty.

OUTPUT

```
constructor called  
Car  created object  
destructor called
```

5. Write a C++ program to create a class for a bank account with a constructor and a destructor.

```
#include <iostream>  
using namespace std;
```

```
class BankAccount{  
public:  
    BankAccount(){  
        cout<<"constructor called"<<endl;  
    }  
  
    ~BankAccount(){  
        cout<<"destructor called"<<endl;  
    }  
};
```

```
int main(){  
    BankAccount b;  
    cout<<"bank account created object"<<endl;  
    return 0;  
}
```

PROGRAM EDITOR

C++

Run

Save

```
1 #include <iostream>
2 using namespace std;
3 class Bank{
4     public:
5     Bank(){
6         cout<<"constructor called"<<endl;
7     }
8     ~Bank(){
9         cout<<"destructor called"<< endl;
10    }
11 };
12
13 int main(){
14     Bank b;
15     cout<<"bank created object"<<endl;
16 }
17
```

INPUT

Your INPUT goes
here!

```
constructor called
bank created object
destructor called
```